

AegisBench: A Physically Calibrated Atmospheric Degradation Benchmark for Aerial Disaster-Response Imagery

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Abstract—Atmospheric scattering and photon-limited illumination, routinely modeled and corrected out elsewhere in remote sensing, are the operating condition in aerial disaster response rather than a nuisance: the event that triggers a flight often fills the air with smoke or dust or leaves responders working after dark. Yet the airborne imagery used to benchmark automated person detection for search and rescue is acquired in clear air and good light. We quantify the resulting gap. AegisBench applies nine corruptions derived from documented radiative transfer and sensor models, including Koschmieder scattering for the smoke and dust families and a linear-domain photometric model for illumination loss, at three severities each calibrated against a declared and machine-checked image statistic. Three architecturally distinct detectors are evaluated across 168 conditions on two airborne datasets spanning the high- and low-altitude regimes, scored by one evaluator at operating points frozen on clear-condition validation data. Degradation is strongly condition-dependent: at the highest severity, rain costs YOLOv11 only 3.6 points of recall on low-altitude imagery but removes almost all of it at high altitude. Illumination loss is the one universal failure. At 4.5 stops of exposure loss, recall reaches exactly 0.000 for all three detectors on both datasets, all 1000 bootstrap resamples agree, and mAP₅₀ falls to at most 0.010, confirming the collapse independently of any confidence threshold. Two corruptions matched to within 3% in mean optical depth differ 6.7-fold in cost, so the loss is not reducible to contrast reduction alone. An unmodified detector run on 300 real night images from an unrelated airborne benchmark recovers 2 of 759 annotated persons, consistent with the modeled result.

Index Terms—Aerial imagery, atmospheric scattering, benchmark, disaster response, low-light imaging, object detection, radiative transfer, robustness, search and rescue, unmanned aerial vehicles.

I. INTRODUCTION

AIRBORNE optical imaging of the land surface is governed by the atmosphere between the sensor and the ground. Scattering and absorption by aerosols attenuate scene radiance and add path radiance, and the available illumination sets the photon flux the sensor integrates. Both effects are routinely modeled in remote sensing, and both are treated as nuisances to be corrected before analysis. When the analysis is automated person detection for land search and rescue, they stop being nuisances and become the operating condition. A search flight is launched because something has gone wrong on the ground, which is frequently the same event that fills the air with smoke or dust, or that leaves responders working after sunset.

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The benchmark code, corruption engine, and evaluation pipeline are available at <https://github.com/Altis-2026/AegisBench>. No imagery is redistributed; every corrupted image regenerates bit-identically from the source datasets.

The imagery used to develop and benchmark aerial person detectors does not reflect this. HERIDAL [1] and SARD [2], the two standard datasets for the task, were both acquired in clear air and good light, and detectors trained on them report strong accuracy under those conditions. What happens to that accuracy when optical depth rises or illumination falls is, at present, not measured. For a capability whose failures are missed rescues, the absence of that measurement is itself the problem.

This letter supplies it. We construct nine corruptions of airborne optical imagery from documented radiative transfer and sensor models rather than from generic image-processing filters, calibrate each to a declared and machine-checked image statistic, and evaluate three architecturally distinct detectors across all resulting conditions on both datasets. Three properties separate this from a transplant of the common-corruptions methodology [3], [4] into aerial imagery. First, the corruption parameters are physical quantities, so a severity level corresponds to a stated optical depth or a stated exposure loss rather than to an arbitrary index. Second, severity is verified against a measured image statistic instead of chosen by inspection. Third, each detector's confidence threshold is fixed once on clear-condition validation data and held across every degraded condition, which is the constraint a deployed system actually operates under.

Robustness benchmarking for airborne imagery under degraded conditions is recent and partial. UAV-C [5] evaluates UAV tracking, not detection, against generic corruptions. HazyDet [6] pairs physics-driven synthetic haze with real hazy drone photographs, covering one condition and general object categories rather than persons. WRRT-DETR [7] builds a weather-robust detector around a maritime dataset. Each addresses a different task, a narrower set of conditions, or a different surface. None measures how aerial search-and-rescue person detection degrades as a function of a physical atmospheric or radiometric parameter, which is what we report.

II. CORRUPTION MODELS

Nine corruptions are grouped into four families by the disaster environment they represent: flood (water_glare, turbidity_cast, inundation), wildfire (smoke_haze, fire_warm_tint), storm (rain_streaks, motion_blur, low_light), and earthquake or post-disaster dust (dust_haze). Table I lists each with its optical mechanism and calibration statistic.

A. Atmospheric Scattering

The three haze corruptions follow the Koschmieder model [8], [9] used throughout atmospheric correction,

$$I(x) = J(x)t(x) + A(1 - t(x)), \quad (1)$$

where J is clear-scene radiance, A is airlight, and t is transmission. Severity is set by the mean transmission $\bar{t}$, so each level corresponds to a mean optical depth $\tau = -\ln \bar{t}$ (Table I). Transmission is spatially varying rather than constant, generated as a band-limited noise field, since haze over terrain is not uniform across a frame.

`smoke_haze` and `dust_haze` differ principally in airlight chromaticity, neutral gray for combustion aerosol against brown for suspended mineral particulate, and in the spatial structure of t . Dust uses twice the transmission variance of smoke (0.30 against 0.15) at a finer base resolution (8 against 3 octave cells), plus low-amplitude granular noise representing near-lens particulates. `fire_warm_tint` applies (1) with a warm airlight and a white-balance shift toward orange, modeling low color-temperature firelight through smoke.

B. Illumination Loss

`low_light` is modeled in the approximately linear photometric domain [10]. The image is linearized with a decode gamma of 2.2, scaled by a photon gain g , shifted toward twilight chromaticity, and returned through the inverse gamma with signal-dependent shot noise and additive read noise whose relative magnitude grows as g falls. Severity is set by g , so each level corresponds to an exposure loss of $\log_2 g$ stops: -1.60 , -2.94 , and -4.47 for severities 1 through 3. This is the parameter against which the headline result is plotted in Fig. 1.

The remaining corruptions are not atmospheric processes. `rain_streaks` follows the directional streak model of Garg and Nayar [11] over a blurred veil, `motion_blur` models wind-induced platform shake, `turbidity_cast` blends toward a mud chromaticity and compresses contrast, `inundation` alpha-blends semi-transparent standing water under a smooth mask, and `water_glare` adds specular sunglint saturation.

C. Calibration and Determinism

Each corruption declares a measurable image statistic whose subset mean must move monotonically across severities, verified on a fixed random 20-image subset of the HERIDAL test split held constant across all corruptions and severities. Eight of nine pass directly. `rain_streaks` does not: streak density has no closed-form image measure, and the substituted edge-strength proxy is confounded, since added streaks raise measured edge strength while the veil blur applied in the same corruption suppresses it. Its severity is instead set by its generation parameters, which increase by construction. Recall still degrades monotonically with it (Fig. 2), independent evidence that the corruption intensifies where this one proxy does not track it.

The calibration statistic makes a severity ladder falsifiable within a corruption. It is not a cross-corruption difficulty

normalizer, and we make no such claim. `smoke_haze` and `dust_haze` illustrate the point: they reach $\tau = 1.386$ and $\tau = 1.427$ at severity 3, matched to within 3%, and nearly identical RMS contrast (0.0475 against 0.0489), yet differ 6.7-fold in recall cost (Section IV).

All pixel-unit parameters are specified per 1000 pixels of the longer image side and scaled at runtime, so a severity means the same thing on a 4000×3000 frame and a 640×640 frame. Corruptions are appearance-only and do not displace object content, so clear-condition ground-truth boxes stay valid; the exception is `inundation`, whose refraction ripple reaches about 1.4 pixels at severity 3 on a 4000-pixel frame, well below person scale. Every stochastic element is seeded by a SHA-256 hash of the image identifier, corruption, severity, and global seed, so the degraded benchmark regenerates bit-identically on any machine.

III. DATA AND EVALUATION PROTOCOL

A. Datasets

HERIDAL [1] is high-altitude orthophoto-scale wilderness imagery at 4000×3000 pixels; we use its official 101-image test split, with 1391 training and 155 validation images carved deterministically from the official training split. For SARD [2], lower-altitude drone video of staged rescue scenarios, we use a public re-export resized to 640×640 (4033 training, 860 validation, 862 test images). SARD frames originate from video, so temporally adjacent frames are near-duplicates. Splitting is group-aware, assigning every image sharing a filename token to exactly one split, and we verified directly that no group appears in more than one. The two datasets bracket the capture regimes in which this task is flown.

B. Detectors and Training

We fine-tune Faster R-CNN [12], a two-stage CNN, YOLOv11 [13], a single-stage CNN, and RT-DETR [14], a transformer detector, from pretrained weights, one run per model per dataset, all at 1024-pixel input and seed 17. Faster R-CNN trains 26 epochs at learning rate 0.005, YOLOv11 80 epochs at 0.01, and RT-DETR 80 epochs at 0.0001, following each architecture's established practice rather than a shared schedule. Degraded data enters neither training nor threshold selection. Training used a single 8-GB consumer GPU, so the benchmark is reproducible without a cluster.

C. Protocol

Each detector's confidence threshold is selected once per dataset by maximizing F1 on clear-condition validation data and frozen across all 28 conditions: 0.99 and 0.96 for Faster R-CNN on HERIDAL and SARD, 0.65 and 0.60 for RT-DETR, 0.45 and 0.30 for YOLOv11. Re-tuning per condition would measure a best case unavailable to a system that must commit to a threshold before a flight. All three architectures are scored by one evaluator using greedy score-ordered matching at IoU ≥ 0.5 , one prediction per ground-truth box, so no detector is scored by its own framework's metric.

Both datasets pass through the same tiled inference pipeline [15] with 1024-pixel tiles and 256-pixel overlap: a HERIDAL frame splits into 20 overlapping tiles, a SARD frame passes through whole. Predictions are mapped back to full-image coordinates and merged with class-agnostic non-maximum suppression, so evaluation is always against full-resolution ground truth. Corruptions are applied before tiling, since haze gradients and flood masks are spatially coherent across a scene. Confidence intervals are bootstrap percentiles over 1000 resamples of the test set by image [16], and every result row logs the commit, a hash of the corruption configuration, the seed, and a timestamp.

IV. RESULTS

A. Clear-Condition Baselines

The top row of Table II gives clear-condition recall. Precision is 0.855 to 0.903 on HERIDAL and 0.955 to 0.972 on SARD, and mAP_{50} is 0.785 to 0.835 and 0.896 to 0.929 respectively. On HERIDAL these sit modestly below the commonly cited YOLOv5L reference of about 0.90 precision and 0.893 recall [1] and are comparable in precision and mAP ; broad agreement across different architectures and input pipelines is the claim, not superiority. Recall is higher on SARD than on HERIDAL, consistent with lower-altitude capture and larger relative object size. Because the two datasets do not share a difficulty floor, degradation is read against each model's own baseline.

B. Degradation Is Condition-Dependent

Table II reports recall at severity 3 for every corruption, detector, and dataset. The spread is the central observation. On SARD, `rain_streaks` costs YOLOv11 3.6 points of recall (0.891 to 0.855) while `low_light` removes all of it. The same corruption behaves differently across capture regimes: `rain_streaks` and `motion_blur` are nearly harmless at low altitude but remove essentially all recall at high altitude, where persons subtend far fewer pixels and a streak or a blur kernel of the same physical scale covers a larger fraction of the target. A single averaged robustness score would hide both effects.

Fig. 2 shows the severity progression for YOLOv11 on SARD. The ordering is not predicted by the calibration statistics. `smoke_haze` and `dust_haze` arrive at severity 3 with mean optical depths within 3% of each other and RMS contrast within 3%, yet recall differs 6.7-fold (0.391 against 0.058). We hypothesize that the difference is carried by the spatial structure of the transmission field rather than by its mean: dust uses twice the transmission variance at a finer base resolution, so it disrupts the high-spatial-frequency content that defines a small aerial target, while smoke mostly lowers global contrast. This is a hypothesis. The present data establish the discrepancy but do not isolate whether airlight chromaticity, transmission granularity, or added grain carries it, which requires an ablation we have not run.

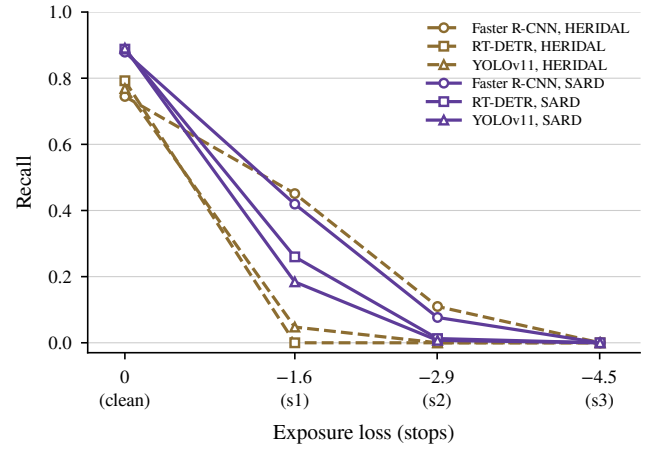


Fig. 1. Recall against exposure loss under `low_light`, plotted against the physical drive parameter rather than a severity index. All six detector and dataset combinations reach exactly zero recall by -4.47 stops. Dashed lines are HERIDAL, solid lines SARD.

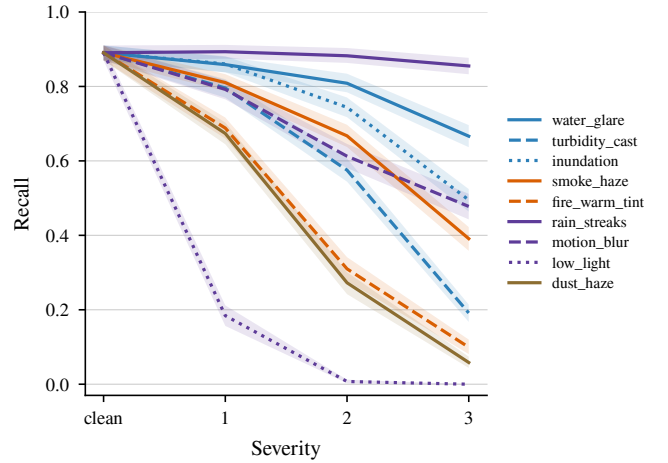


Fig. 2. Recall against severity for all nine corruptions, YOLOv11 on SARD, with bootstrap 95% intervals shaded. Colour indicates disaster family, line style distinguishes corruptions within a family. All curves begin at the same clear-condition baseline.

C. Illumination Loss Produces Total Failure

`low_light` is the one corruption that behaves the same way everywhere. Table III and Fig. 1 give the result: by -4.47 stops, recall is exactly 0.000 for all three detectors on both datasets, and all 1000 bootstrap resamples agree, so the interval is degenerate at $[0.000, 0.000]$. A two-stage CNN, a single-stage CNN, and a transformer detector are about as architecturally different as detectors get, and all three sit at the same floor, which rules out an architecture-specific explanation.

A frozen-threshold zero could in principle reflect a badly placed operating point rather than absent detections. mAP_{50} settles this, because it scans every threshold. It collapses in step with recall (Table III): across all six detector and dataset pairs at severity 3, mAP_{50} is at most 0.010, against clear-condition values of 0.785 to 0.929. If this were a calibration artifact, mAP would stay high while recall fell. It does not, so the detections are absent rather than merely below threshold.

TABLE I
THE NINE CORRUPTIONS, THEIR OPTICAL MECHANISM, AND THE STATISTIC DEFINING EACH SEVERITY LADDER

Corruption	Family	Optical mechanism	Calibration statistic	s1	s2	s3
water_glare	Flood	Specular sun-glint saturation	saturated fraction $\uparrow$	0.0116	0.0182	0.1895
turbidity_cast	Flood	Sediment-laden water, contrast loss	RMS contrast $\downarrow$	0.1098	0.0701	0.0378
inundation	Flood	Semi-transparent standing water	occluded fraction $\uparrow$	0.0830	0.2235	0.3921
smoke_haze	Wildfire	Koschmieder, gray airlight (τ 0.36/0.80/1.39)	RMS contrast $\downarrow$	0.1170	0.0776	0.0475
fire_warm_tint	Wildfire	Low-CCT firelight through warm haze	red/blue ratio $\uparrow$	1.6743	2.0721	2.4516
rain_streaks	Storm	Directional streaks over a rain veil	generation parameters [†]	—	—	—
motion_blur	Storm	Wind-induced platform shake	edge strength $\downarrow$	0.0764	0.0572	0.0472
low_light	Storm	Photon scaling ($-1.6/ -2.9/ -4.5$ stops)	mean luminance $\downarrow$	0.2355	0.1533	0.1175
dust_haze	Earthquake	Koschmieder, brown airlight (τ 0.43/0.87/1.43)	RMS contrast $\downarrow$	0.1094	0.0720	0.0489

Arrows give the required direction as severity increases. Values are means over a fixed random 20-image subsample of the HERIDAL test split, held constant across all corruptions and severities, and monotonicity is checked automatically on these means. Mean optical depth $\tau = -\ln \bar{t}$ and exposure loss $\log_2 g$, given in parentheses for the three severities, are derived from the declared generation parameters rather than measured from the imagery.

[†] *rain_streaks* has no closed-form image statistic isolating streak density, so its severity is set by its generation parameters (streaks per megapixel, streak length, veil intensity, all increasing by construction).

TABLE II
RECALL AT SEVERITY 3 FOR EVERY CORRUPTION, DETECTOR, AND DATASET

Corruption	Family	HERIDAL ($n = 101$)			SARD ($n = 862$)		
		Faster R-CNN	RT-DETR	YOLOv11	Faster R-CNN	RT-DETR	YOLOv11
<i>clear condition</i>	—	0.745	0.792	0.769	0.879	0.888	0.891
water_glare	Flood	0.614	0.682	0.653	0.629	0.646	0.666
turbidity_cast	Flood	0.024	0.042	0.163	0.180	0.074	0.191
inundation	Flood	0.326	0.472	0.398	0.489	0.459	0.495
smoke_haze	Wildfire	0.148	0.320	0.407	0.303	0.319	0.391
fire_warm_tint	Wildfire	0.401	0.086	0.145	0.407	0.118	0.099
rain_streaks	Storm	0.003	0.003	0.056	0.778	0.840	0.855
motion_blur	Storm	0.000	0.000	0.003	0.398	0.329	0.478
low_light	Storm	0.000	0.000	0.000	0.000	0.000	0.000
dust_haze	Earthquake	0.098	0.071	0.104	0.160	0.050	0.058

Recall at each model's frozen operating point, selected on clear-condition validation data only. *low_light* is the sole corruption reaching exactly zero for every detector on both datasets. *rain_streaks* and *motion_blur* separate the two capture regimes most sharply, costing little at low altitude and nearly all recall at high altitude.

Two statements hold on both datasets. Total failure by -4.47 stops replicates for all three architectures, and Faster R-CNN is consistently the most robust at every severity where any model retains recall, despite carrying the strictest threshold (0.99 on HERIDAL against 0.65 and 0.45), which rules out threshold strictness as the explanation. One statement does not hold: RT-DETR loses all recall at every severity on HERIDAL, including the mildest, but outperforms YOLOv11 at severities 1 and 2 on SARD. That ordering is dataset-dependent, not an architecture property, and we report it as such.

The failure is missed detections, not displaced ones. Measuring IoU between the boxes predicted for instances detected in both the clear and degraded conditions [17], this stability stays between 0.779 and 0.984 for YOLOv11 on SARD across all 27 degraded conditions while recall sweeps from 0.891 to 0.000, and the count of commonly detected instances tracks recall exactly (202, then 8, then 0 across the three *low_light* severities). Both decline, and we do not claim independence, but the magnitudes differ sharply: surviving detections stay on the person, so a mitigation should target sensitivity rather than localization.

D. Real Night Imagery

To test the modeled result outside our own pipeline, we ran the SARD-trained YOLOv11 model, unmodified, on 300 real night images from VisDrone [18], an unrelated airborne benchmark. Images were selected as the lowest-luminance frames in its test split and manually confirmed to show genuine dusk or night scenes. Recall against VisDrone's own person annotations was 0.003, recovering 2 of 759 annotated persons, consistent with the total failure the modeled corruption reaches at severity 3. The low-light-mitigated checkpoint recovers 4 of 759 on the same images, moving the floor as it does on the modeled corruption but not to anything an operator could rely on. This remains a sample from a different surface type, lit urban scenes rather than wilderness terrain, and it is not a substitute for systematic validation against real adverse-condition search imagery; it is, however, evidence from real sensor data that the failure is not an artifact of the corruption model.

V. LIMITATIONS

The corruptions are modeled and applied to real imagery, following established practice for controlled degradation studies [3], [19]. That design is what makes per-condition attri-

TABLE III
DETECTION UNDER ILLUMINATION LOSS: RECALL AT THE FROZEN OPERATING POINT AND THRESHOLD-INDEPENDENT MAP₅₀

Model	Dataset	Thr.	Recall [bootstrap 95% interval]			mAP ₅₀		
			−1.60 stops	−2.94 stops	−4.47 stops	−1.60	−2.94	−4.47
Faster R-CNN	HERIDAL	0.99	0.451 [0.365, 0.540]	0.110 [0.064, 0.163]	0.000 [0.000, 0.000]	0.537	0.176	0.010
RT-DETR	HERIDAL	0.65	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.001	0.000	0.000
YOLOv11	HERIDAL	0.45	0.048 [0.027, 0.073]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.064	0.000	0.000
Faster R-CNN	SARD	0.96	0.419 [0.389, 0.452]	0.077 [0.060, 0.094]	0.000 [0.000, 0.000]	0.546	0.142	0.000
RT-DETR	SARD	0.60	0.260 [0.233, 0.288]	0.013 [0.007, 0.019]	0.000 [0.000, 0.000]	0.421	0.049	0.001
YOLOv11	SARD	0.30	0.184 [0.157, 0.212]	0.007 [0.003, 0.012]	0.000 [0.000, 0.000]	0.247	0.018	0.000

Exposure loss in stops is $\log_2 g$ for the declared photon gain g , corresponding to severities 1 through 3. Recall and mAP₅₀ are point estimates on the full test set; bracketed intervals are bootstrap percentiles over 1000 resamples by image. Every interval at −4.47 stops is degenerate at zero, meaning all 1000 resamples returned zero recall. mAP₅₀ scans all confidence thresholds and so is immune to the objection that a frozen-threshold zero reflects a badly placed operating point; it collapses in step, reaching at most 0.010 at −4.47 stops against clear-condition values of 0.785 to 0.929.

bution possible, since real adverse-condition imagery rarely comes with a matched clear counterpart of the same scene, but it is also the ceiling: the benchmark measures response to modeled optics, and the VisDrone check is a spot check rather than systematic validation. We train one seed per model per dataset, so bootstrap intervals quantify test-set variance, not training variance; a three-seed repeat and a full mitigation study, including its cost on the other eight corruptions, appear in a companion submission on the same pipeline. The mitigated checkpoint above is one arm of that study.

VI. CONCLUSION

Measuring degradation against physical parameters rather than an arbitrary severity index turns “the system is not robust” into a specific, falsifiable claim tied to an operating condition a responder can recognize. It also carries a consequence for how a system should be selected and deployed: a robustness figure measured at one altitude regime does not transfer to another, an illumination collapse this uniform across architectures is unlikely to be fixed by choosing a different detector, and a corruption’s cost is not predictable from its effect on a generic image statistic. None of these three findings would survive being compressed into a single averaged robustness score, which is the form most benchmarking practice still reports in.

Two directions follow. Ablating transmission-field granularity against airlight chromaticity would establish whether spatial frequency or colour carries the smoke and dust discrepancy. Systematic validation against real adverse-condition search imagery remains the largest open gap, and the hardest, because such imagery is rarely collected and released.

ACKNOWLEDGMENT

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